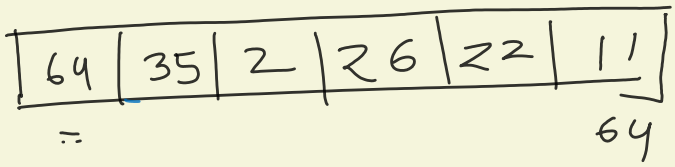
Sorting :-

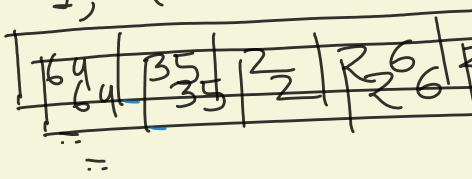
① Selection Sort :-



$i = 0, n$

$j = i+1, n$

- Find the smallest element and place it in its correct position
- The smallest element gets sorted 1st

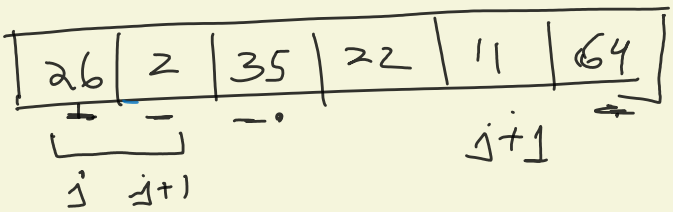


② Bubble Sort :-

- it checks within a bubble, compares current and the current +1and moves the bigger one
- Largest one will be sorted 1st

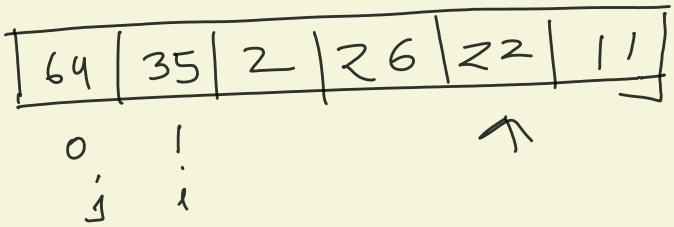
$i = 0, 1, n$

$j = 0, n-1-i$

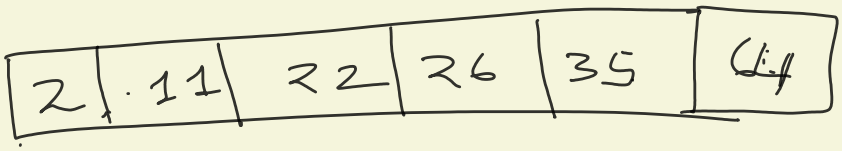


③ Insertion Sort :-

- Two arrays need to maintain.
  - sorted
  - unsorted
- Pick one item at time and place it in the right place in the sorted array

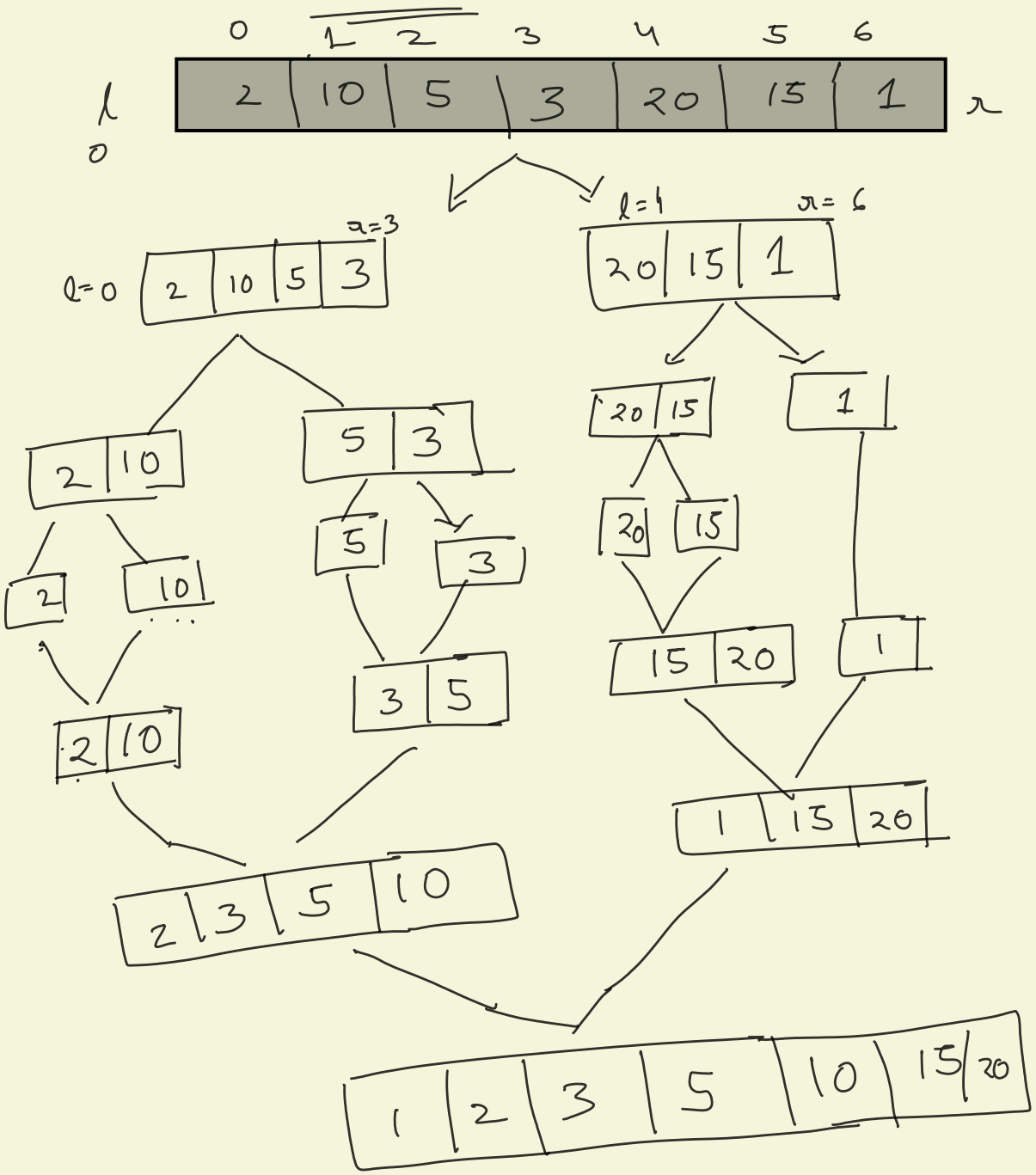


sorted item = 35



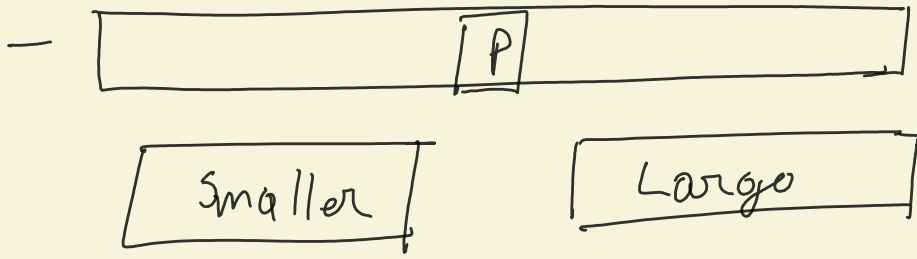

```
for i in range (1, n)
    temp = arr[i]
    j = i-1
    while (j >= 0 and
           temp < arr[j]
        a[j+1] = a[j]
        j -= 1
    arr[j+1] = temp
```

4. Merge Sort :- Divide & Conquer

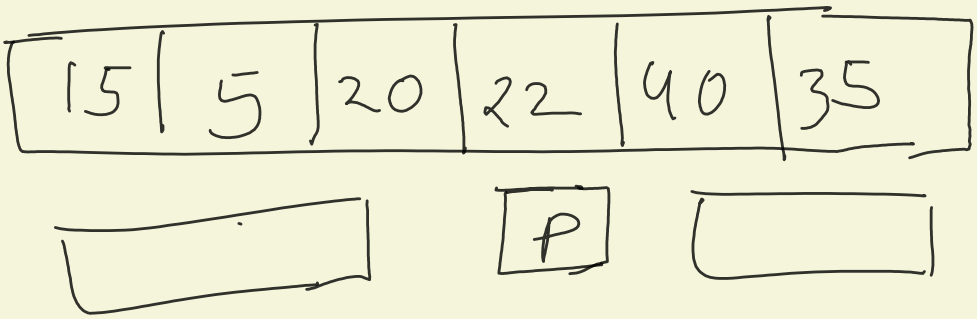


5. Quick Sort :- Divide & Conquer

- Pivot



Example



pivot = 22

```
i < pivot
    i++
j > pivot
    j--
if i<j :-
  swap(i,j)
```